

AquaShop — Architecture

A polyglot Kubernetes platform for a freshwater aquarium shop.
Designed for AWS · validated with `terraform test` · run on k3d.

READ THIS BEFORE ANYTHING ELSE

The AWS infrastructure layer in this repository has **never been applied to an AWS account**. It is written as production-quality Terraform and validated in CI with `fmt`, `validate`, `tflint`, `checkov` and `terraform test` against mock providers. That proves module composition, variable contracts and policy compliance. It does not prove AWS behaviour.

The runtime is a single-node **k3d** cluster inside WSL2 on a 16 GB laptop. Every diagram in this document carries AWS labels because AWS is the target platform, and every place where the local setup diverges from that target is named in the local-to-cloud section rather than glossed over.

The short version, said out loud without hedging: **"I built it AWS-first and ran it locally because I was paying for it myself."**

PHASES 1–5 COMPLETE

ALL 8 SERVICES RUN IN k3d

STOREFRONT REBUILT 28 SEP

253 PRODUCTS (V26)

PHASES 6–7 PLANNED

AWS: VALIDATED, NOT APPLIED

I The product, and why this domain

AquaShop sells live freshwater fish, shrimp, plants, hardscape, equipment and food, with a full care profile published for every living species before purchase. The domain was chosen because live livestock imposes constraints a generic shop demo does not have, and each constraint forces an architectural decision that is worth defending in an interview.

DOMAIN CONSTRAINT → ARCHITECTURAL CONSEQUENCE	
Perishable stock	Livestock ships only inside a safe weather window, so an order can be <i>confirmed but not yet shippable</i> . The order state machine has a waiting state that a normal shop does not, and it is time-driven rather than event-driven.
Stock per tank	Stock is held per physical tank, not as one integer. Reservations are tank-scoped, time-bounded and idempotent, which is a distributed-systems problem (TTL expiry, retry safety) rather than a decrement.
Compatibility	Species are not mutually compatible. That is a rules engine over water parameters and temperament, not CRUD — which is what justifies a separate service with a different language and a different change cadence.
Water parameters	pH, dGH and temperature ranges must <i>overlap</i> across every inhabitant. Interval intersection over a set, evaluated against tank volume.
DOA claims	High-value livestock sometimes arrives dead, so there is a claims workflow with photo evidence and a compensating refund path — a real saga compensation, not a contrived one.

EIGHT SERVICES, AND THE PROPERTY THAT JUSTIFIES EACH LANGUAGE		
SERVICE	STACK	WHY THIS STACK
catalog-service	Java 21 / Spring Boot	Read-heavy relational data with a rich schema; the ecosystem I know best, so the platform work is the novel part.
order-service	Java 21 / Spring Boot	Long-lived transactional orchestration; mature saga and scheduling libraries.
inventory-service	Go	Many small concurrent hold/expire operations; a small footprint suits a service that scales on request count.
payment-service	Rust / Axum	Smallest surface, strictest correctness: exhaustive compile-time matching, non-wrapping arithmetic. Cost: slowest CI, smallest maintainer pool.
aquatics-advisor	Python / FastAPI	Rules and range logic that changes often, edited by domain people, not platform people.
notification-service	Go	Planned, not built. I/O-bound fan-out to email and webhooks.
staff-portal	JSP / WildFly	Planned, not built. A slow-starting WAR — the exercise that connects this to production work.
storefront	TypeScript / Fastify	A BFF is I/O aggregation; server-rendered HTML avoids a client state layer that proves nothing.

System architecture

AquaShop System Architecture

All eight services are built and run together in k3d. Messaging, observability and GitOps remain planned.

Designed for AWS • validated with terraform test and mock providers • run on k3d • never applied

I Deployment topology

Deployment Topology

What actually runs, beside what it is designed to become.

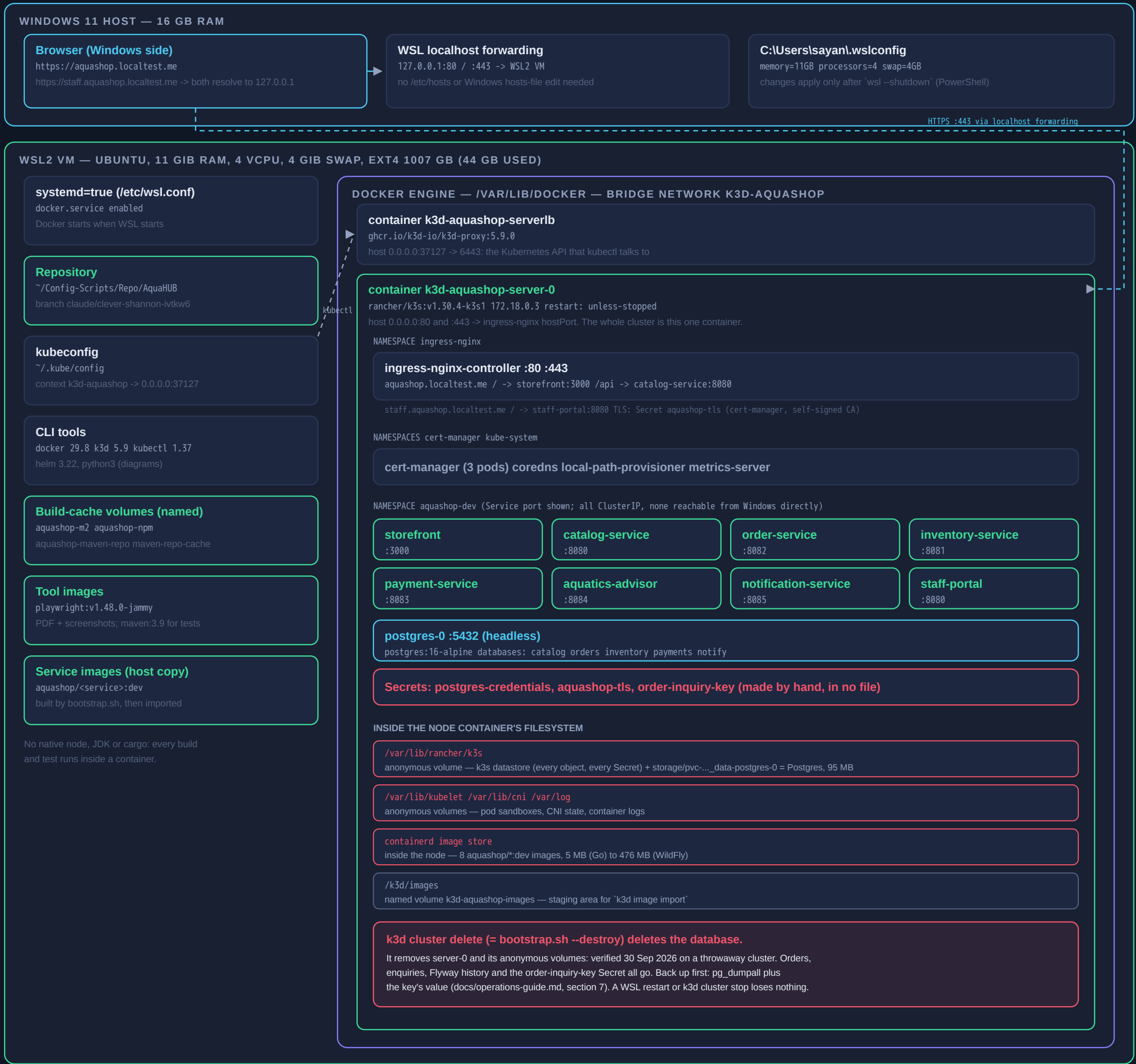
Physical deployment — what runs where on this machine

Physical Deployment — this machine

Where every process, port and byte of data actually lives. Read off the live machine, 30 Sep 2026.

SURVIVES --destroy

LOST ON --destroy



Survives a WSL restart: everything (restart: unless-stopped). Survives --destroy: only the repo, the host images and the named volumes.

AquaShop / docs/diagrams/physical.svg

I Operating it without Claude — the one-page version

The full procedure, with every troubleshooting path, is docs/operations-guide.md; installing from a bare Windows machine is docs/installation-manual.md. This page is what to remember.

START AND CHECK (AFTER A REBOOT, NOTHING TO DO)

systemd starts Docker when WSL starts, and both k3d containers are restart:unless-stopped, so the cluster comes back by itself.

```
docker ps
kubectl get nodes
kubectl -n aquashop-dev get pods
curl -sk https://aquashop.localtest.me/healthz
```

UPDATE ONE SERVICE AFTER GIT PULL

```
docker build -t aquashop/storefront:dev services/storefront
k3d image import -c aquashop aquashop/storefront:dev
kubectl -n aquashop-dev rollout restart deploy/storefront
kubectl -n aquashop-dev rollout status deploy/storefront
```

Or rebuild everything: ./scripts/bootstrap.sh --profile full-app --metrics. It reuses the cluster and keeps the data.

WHEN IT WILL NOT COME UP: LOOK IN THIS ORDER

1 Docker	docker info → sudo systemctl start docker
2 Cluster	k3d cluster list → k3d cluster start aquashop
3 Pods	kubectl -n aquashop-dev get pods
4 Why	kubectl -n aquashop-dev describe pod <pod> kubectl -n aquashop-dev get events --sort-by=.lastTimestamp
5 Logs	kubectl -n aquashop-dev logs deploy/<svc> --previous
6 Edge	kubectl -n ingress-nginx logs deploy/ingress-nginx-controller

BEFORE YOU EVER RUN --DESTROY

The database and the enquiry key live inside the k3d node container. Back them up first:
kubectl -n aquashop-dev exec postgres-0 -- pg_dumpall -U postgres --clean --if-exists > ~/aquashop.sql
kubectl -n aquashop-dev get secret order-inquiry-key -o jsonpath='{.data.INQUIRY_ENCRYPTION_KEY}' | base64 -d > ~/aquashop-inquiry-key

Delivery Flow — Two Repositories

Nobody runs kubectl apply. Promotion is a pull request.

DEVELOPER

feature branch
git push

AQUASHOP (APP REPO)

Pull request
review + required checks

merge to main
squash, one commit

GITHUB ACTIONS

lint

unit + IT

coverage gate

Trivy scan

buildx

push :<sha>

PR to GitOps repo

Image tag is the git SHA. Never :latest — a mutable tag makes rollback a guess and makes 'which image is running' unanswerable from the cluster.

AQUASHOP-PLATFORM (GITOPS REPO)

dev overlay
image tag bumped automatically

uat overlay
promotion PR, human approval

prod overlay
promotion PR, human approval

The GitOps commit log is the literal deployment history: every change to what runs is a commit, and a rollback is `git revert` of that commit, not a command typed against a cluster.

CLUSTER

Argo CD app-of-apps
auto-sync + self-heal on dev

aquashop-dev namespace
rolling update, maxUnavailable 0

uat / prod namespaces
replicas: 0 until materialised

INTERVIEW HOOKS

Why two repos: a CI bot writing to the app repo would retrigger CI on its own commit.

Drift: Argo self-heal reverts a manual kubectl edit within seconds; that is the point, and it is also how you lock yourself out of an emergency fix.

Rollback: revert the tag-bump commit.
Mean time to recover = sync interval.

Definition of done for this flow: a merged pull request deploys a new version with no manual kubectl anywhere.

AquaShop / docs/diagrams/delivery-flow.svg

I The memory budget — the constraint that shaped the design

16 GB of physical RAM on the laptop. The design targets 11 GB usable inside WSL2 via a `.wslconfig` memory override. It is applied: `/proc/meminfo` measures **11 GiB** (it measured 7.4 GB, WSL2's unconfigured default, until some point before 22 Sep 2026). The full platform does not fit at once, so the cluster is built as **profiles**: named subsets brought up for a purpose. This is not a workaround bolted on at the end; it is why NATS replaced Kafka, why one Postgres instance hosts per-service databases, and why only one environment is materialised at a time.

PROFILES		
PROFILE	ADDS	TOTAL
core	k3d, ingress-nginx, cert-manager, Postgres, catalog, storefront	1.32 GiB measured
commerce	order, inventory, payment, advisor, NATS	~2.05 GiB measured*
full-app	notification, staff-portal (WildFly)	2053 MiB measured
platform	Argo CD	~6.1 GB estimate
observability	kube-prometheus-stack, OTel collector, Tempo, prometheus-adapter	~9.2 GB estimate
*NATS not deployed yet by <code>bootstrap.sh --profile commerce</code> (planned for Phase 6); the commerce figure is the four services alone. core+commerce came in well under their old estimates — core at 1.32 GiB against a ~2.6 GB estimate, commerce adding ~730 MiB against a ~4.2 GB additive estimate.		
Rule enforced by the bootstrap script: never build images while the observability profile is up. ~1.8 GB of headroom (against the intended 11 GB ceiling) does not survive a Maven or Cargo build, and the failure mode is the kernel OOM killer taking an unrelated pod.		

DECISIONS THIS CONSTRAINT FORCED, AND WHAT EACH COSTS	
NATS JetStream instead of Kafka	No consumer in this system needs log replay or partition ordering, and Kafka costs ~1 GB this machine does not have. Cost: no partition-key ordering, a smaller tooling ecosystem, and MSK is a heavier migration than the mapping suggests.
One Postgres, database per service	Separate databases and login roles, each with CONNECT on its own database only. Saves ~1.1 GB. Cost: blast-radius isolation is lost — one restart takes every service down, which per-service RDS instances would not. Credential isolation and the no-shared-schema rule are unaffected.
One environment materialised at a time	dev, uat and prod are designed as Argo CD Applications with two at replicas: 0. Only dev exists today, and Argo CD is not built. Cost: no environment has ever run concurrently with another, so cross-environment concerns are untested. Say "three overlays under GitOps, one materialised", never "I ran prod".
No CPU limits on JVM services	A CPU limit means CFS throttling: once the container exhausts its quota it is stopped for the remainder of each 100 ms period, which appears as latency spikes while the node looks idle. Memory is limited because memory is not compressible. Cost: a runaway pod can starve its neighbours; the ResourceQuota is the backstop.

A CEILING THAT WAS MEASURED LATE

For the first week the WSL2 ceiling was the **unconfigured 7.4 GB**, and the observability estimate (~9.2 GB) could not have fitted at all. That was invisible while every figure here was an estimate. The `.wslconfig` override is now applied and measures 11 GiB, so observability fits *on paper*. It has never been built, so that is still an estimate against a measurement, not a result.

I Local to cloud — Phase 1 components

The full mapping document covers every component, the Terraform that would provision it, and the traffic path end to end. This page is the Phase 1 extract: what is faithful, and what I have therefore *not* validated.

LOCAL	AWS COUNTERPART	TERRAFORM MODULE → CONSUMER	FAITHFUL	NOT FAITHFUL — AND WHAT THAT MEANS
k3d node container	EKS managed node group	eks → outputs cluster_endpoint, oidc_provider_arn; consumed by iam and the addons module	Kubernetes API behaviour, scheduling, probes, quotas, RBAC	Single node, so PodDisruptionBudgets, anti-affinity and node drains are configured but never exercised. Pod IPs come from a Docker bridge, not from VPC subnets, so VPC CNI IP exhaustion cannot occur here .
ingress-nginx hostPort	ALB + AWS Load Balancer Controller	network → public_subnet_ids; consumed by the controller through subnet discovery tags	Ingress objects, path routing, ingress class, nginx timeouts	No target groups, no cross-zone balancing, no deregistration delay. A missing kubernetes.io/role/elb tag silently produces no load balancer — I can explain it, I have not hit it.
cert-manager self-signed CA	ACM certificate on the ALB	acm → certificate_arn; consumed by the Ingress annotation	Certificate lifecycle, secret mounting, renewal mechanics	TLS terminates inside the cluster here and at the load balancer in the target. Cipher policy, SNI and WAF association are untested.
Postgres StatefulSet + local-path PVC	RDS PostgreSQL, Multi-AZ	rds → endpoint, port, secret_arn; consumed by the app Helm values and IRSA policy	Schema, Flyway migrations, connection pooling, credential isolation per service	No failover, no automated backup, no parameter groups. The volume is node-local, so the "volume follows the pod" property of EBS does not exist. Multi-AZ failover is a DNS flip that pools must survive — explainable, not reproducible.
Plaintext Secret in Git	Secrets Manager read via IRSA	iam → irsa_role_arn; consumed by the ServiceAccount annotation	Nothing yet — this is a Phase 1 placeholder	The single largest gap in the repository today. Phase 5 replaces it with External Secrets + SOPS. Until then this is stated as a known limitation, not presented as a design.

Local image import into k3d

ECR with immutable tags ecr → repository_urls; consumed by SHA-tagged images, no :latest, CI and the deployment image field scan-on-build via Trivy

No registry authentication, no pull-through cache, no VPC endpoint for private pulls, no lifecycle policy for untagged images.

I Phases 2–5 — what each service is built on

PHASE 2 · INVENTORY-SERVICE (GO)

Stock held per physical tank, not one integer per SKU. Availability = `on_hand - Σ holds WHERE expires_at > now()` — the deadline is in the query, so the reaper is bookkeeping, not load-bearing.

Locks are taken in one statement, availability read in a second: under READ COMMITTED a single combined statement reads a stale snapshot and oversells — the concurrency test (20 goroutines, 8 fish, 2 at a time) caught exactly that.

Measured locally: 13.9 MiB idle, 17.0 MiB after 200 reservations, 2.5 ms mean, zero oversells across 300 concurrent reservations. **Now also run in k3d: 3 MiB resident** — idle during the smoke test, not under sustained load.

PHASE 4 · PAYMENT-SERVICE (RUST / AXUM)

Smallest surface, strictest correctness: every ledger transition is an exhaustive match with no catch-all arm. The unknown — acquirer took the money and never answered — is a state in both services, never collapsed into success or failure. The intent row is written *before* the acquirer is called, so a mid-call death still leaves a findable record.

Measured: 6 MiB resident locally, **2 MiB in k3d**, against 361/224 MiB for order-service doing comparable work. **18 gated database tests** (cargo test, needs PAYMENTS_TEST_DSN) now cover its CHECK constraints, append-only trigger and race-losing conditional update — previously exercised only by hand. The acquirer itself is still a stub: no partial captures, no chargebacks, no 3-D Secure, no settlement.

PHASE 3 · ORDER-SERVICE (JAVA, THE CHECKOUT SAGA)

Four steps, no distributed transaction: `reserve → authorise → commit → confirm`, each with its own compensation. Stock is held *before* money is taken — a declined card costs a released hold; the reverse ordering costs a refund and an apology. There is no `PAID → PAYMENT_FAILED` transition: the worst outcome is unreachable, not just unlikely.

~~Not proven: the saga is not crash-safe.~~ **Closed.** SagaRecovery scans for orders stuck in PAID/STOCK_RESERVED and finishes them — demonstrated with a real kill -9 mid-checkout. A state scan, not an outbox (ADR 0018, amends 0015).

Live in k3d, 16 Sep 2026: a full checkout through the ingress-internal saga — reserve, authorise, commit, confirm — ended CONFIRMED with a correct dispatch window. Talks to the real payment-service over HTTP; the stub gateway was retired at Phase 4.

PHASE 5 · AQUATICS-ADVISOR (PYTHON / FASTAPI)

The rules change weekly, the code rarely — so thresholds live in `rules.yaml` with a plain-English justification per line, quoted back to the customer. It owns no species data of its own; a local copy would drift the first time somebody corrects a pH range.

Three verdicts, not two: a refusal (no overlap), a caution (narrow overlap), or a pass. **Not proven:** nothing tests the HTTP layer or the catalog client, and the image is `python:3.11-slim` rather than distroless.

I The first k3d run — three defects only a real build could find

Every prior session verified this repository against a local Postgres, never in a cluster, because no session before 16 September 2026 had a Docker daemon. This one did. **core** came up first, **commerce** followed the same day, and a live checkout ran the full saga in-cluster. None of the three defects below was caught by review — each only exists once the actual container runs.

1. EVERY POD · CREATECONTAINERCONFIGERROR

All six deployments sat in `CreateContainerConfigError` on core's first rollout. The distroless `:nonroot` images set `USER nonroot` — a name, not a UID — and `runAsNonRoot: true` alone gives kubelet nothing numeric to verify without running the container first. **Fix:** `runAsUser: 65532 / runAsGroup: 65532` added explicitly to every pod `securityContext`.

2. INVENTORY-SERVICE · TOOLCHAIN DRIFT

`go.mod` declared `go 1.25.0`; the `Dockerfile` pinned `golang:1.24-bookworm`. `go mod download` refused to run. **Fix:** bumped to `golang:1.25-bookworm`.

3. PAYMENT-SERVICE · TOOLCHAIN DRIFT

`Cargo.lock` resolved `home v0.5.12`, whose manifest needs Cargo's `edition2024` feature (stabilized in Rust 1.85) — stricter than the crate's own declared `rust-version = "1.82"`, which the pinned `rust:1.82-bookworm` builder matched but could not satisfy. **Fix:** bumped to `rust:1.90-bookworm`.

`mvn/go build/cargo build` run locally against whatever toolchain happens to be installed, so neither toolchain defect had ever been visible before — only `docker build` against the pinned image actually enforces the version a `Dockerfile` claims.

A LIVE CHECKOUT, END TO END, IN-CLUSTER

From a temporary `curlimages/curl` pod inside the namespace (order-service is intentionally not exposed through the public ingress): create a cart, add a line (INV-AMA-01, Amano Shrimp × 2), checkout with an `Idempotency-Key`.

Result	201
Order state	CONFIRMED
Reservation	COMMITTED
Dispatch	2026-09-21T14:00+05:30
Audit trail	PENDING → STOCK_RESERVED → PAID → CONFIRMED

Dispatch window is the next Monday after the Thursday cut-off — the shipping-calendar rule working against a real clock. `inventory-service` correctly reflected the sale afterward (`held: 0`, `onHand` reduced).

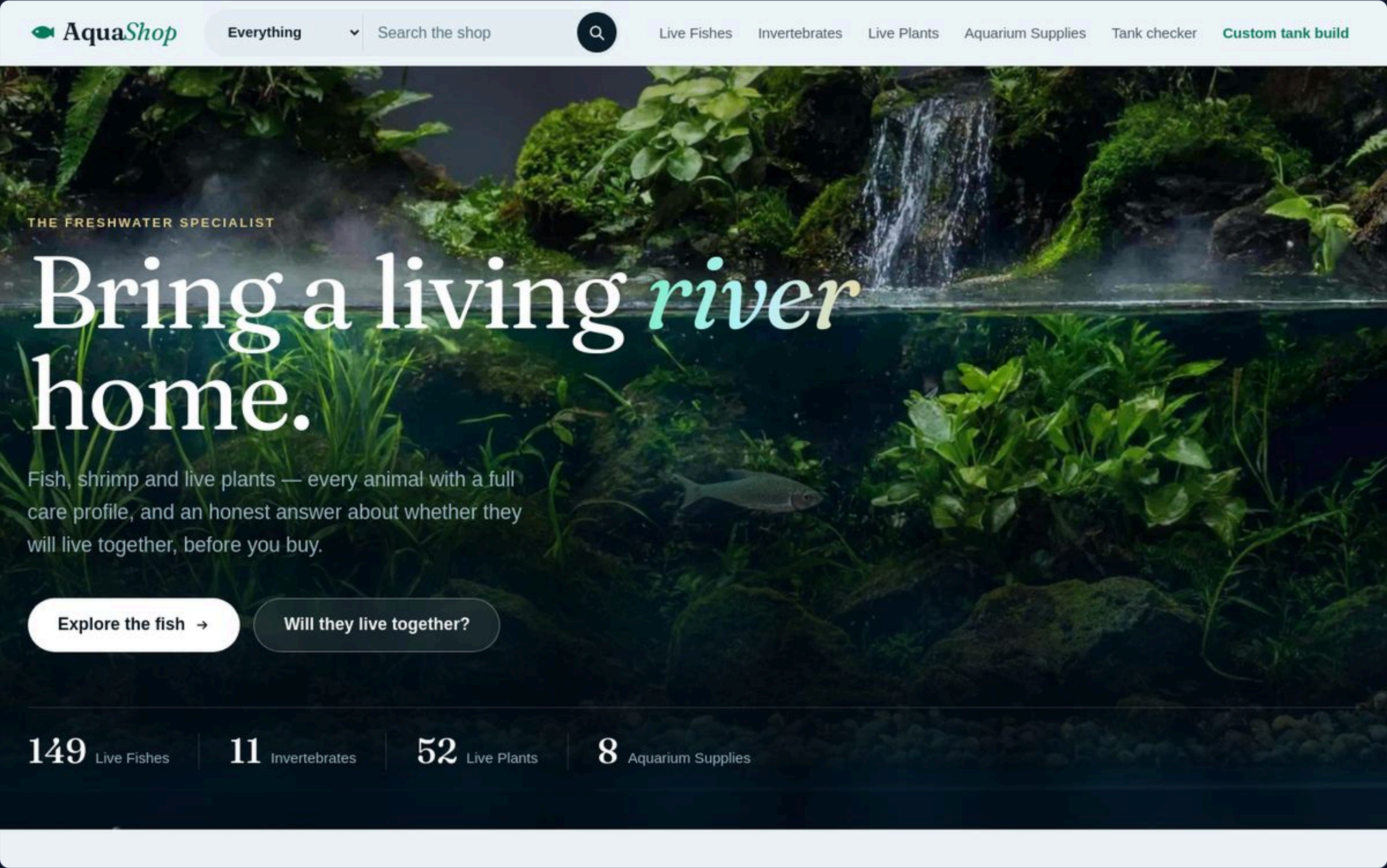
PER-POD, MEASURED (KUBECTL TOP, BOTH PROFILES SETTLED)

catalog-service	JVM, 640Mi limit	215 MiB
order-service	JVM	224 MiB
storefront	Node	30 MiB
aquatics-advisor	Python/FastAPI	43 MiB
inventory-service	Go	3 MiB
payment-service	Rust	2 MiB
postgres		59 MiB
ingress-nginx		184–189 MiB

215/224 MiB are the first honest in-cgroup JVM numbers in this repository (`MaxRAMPercentage=70`, not host RAM). The two smallest figures reflect idle pods during a smoke test, not sustained load.

The storefront — the customer-facing layer (28 Sep 2026)

Rebuilt to make a visitor want to keep fish, without giving up ADR 0005: still one server-rendered HTML page per URL, still working with JavaScript off. What changed is what it looks like, what it sends, and one feature no other fish shop has.



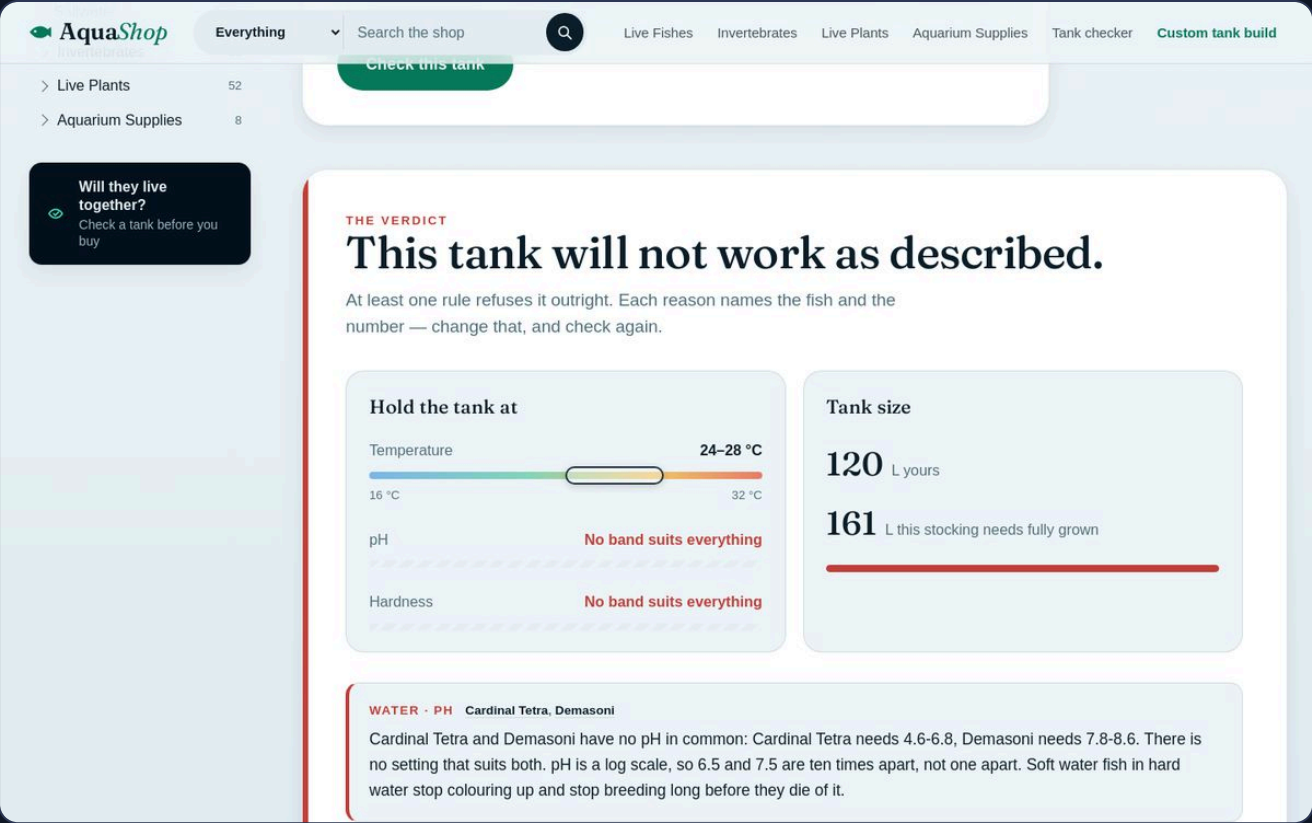
THE TANK CHECKER — AQUATICS-ADVISOR'S FIRST CUSTOMER-FACING DOOR

A customer types a tank size and the fish they want; the storefront resolves names to SKUs and asks the advisor. The verdict names the fish and quotes the shop's own rule for every no, and shows the water band the whole tank can live in. A GET form, so a checked tank is a link.

Verified with the advisor scaled to zero: an honest 502 in 16 ms, the customer's tank kept in the form, every other page unaffected.

BYTES ON THE WIRE, FIRST VISIT, MEASURED

PAGE	BEFORE	AFTER
Home, desktop	4,846 KB	760 KB
Lake Malawi, desktop	7,094 KB	648 KB
Product, desktop	5,162 KB	229 KB
Home, phone 3x	1,979 KB	986 KB



Product,
phone 3×

243 KB

374 KB

Old and new builds side by side against the same catalog, headless Chromium, scrolled to the end. The phone product page grew: it now shows related fish and loads the display font. Repeat visits fell from ~40 KB of revalidation to ~8 KB.

HOW

- **Brotli/gzip** on HTML: the shop front is 58 KB raw, 10 KB sent.
- **WebP variants** (480/960 px) built at image-build time, served by srcset. Cards had been loading 1600 px JPEGs.
- **Cache policy**: CSS and fonts immutable behind a content hash; photos a day (they get replaced under the same name).
- **Hover previews** that really do wait for a hover: the old ones loaded 15 photos per desktop visit.
- **View transitions and speculation rules**: declarative, no code of ours (ADR 0022).

| **Status — what's built, what's measured, what's next**

BUILT AND EXERCISED IN K3D (16 – 30 SEP 2026)

- **All eight services** up together as full-app: 2234 MiB on 17 Sep, 2053 MiB on 30 Sep. notification-service (Go, 4 MiB) and staff-portal (WildFly, 381 MiB, its own ingress host) included.
- **A catalogue of 253 products** across 51 sections (Flyway at V26), with care profiles for every animal and plant. 226 have a licensed photograph; 27 show a placeholder on purpose.
- **Custom tank enquiries**, encrypted by Postgres (ADR 0021), and a **tank checker** on the shop front in front of aquatics-advisor.
- **Probes, resource limits, ingress/TLS** exercised in-cluster for the first time.
- **A live checkout saga** across three services, ending CONFIRMED with a committed reservation and a correct dispatch window.
- **Crash recovery** (SagaRecovery, ADR 0018) demonstrated with a real kill -9 — outside k3d, not yet reproduced against the in-cluster saga specifically.
- **payment-service's** constraints, trigger and race-losing update now covered by 18 gated database tests.

NEXT PHASES

Phase 6	Observability (OTel, RED + business metrics, burn-rate alerts) and the delivery layer (Argo CD app-of-apps across dev/uat/prod). NATS arrives, and notification-service moves from ADR 0020's fire-and-forget HTTP push to a real subscription. Saga crash-recovery is already closed without an outbox (ADR 0018); an outbox still arrives here, but for reliably publishing domain events to NATS, not for crash-safety.
Phase 7	The full local-to-cloud document (only the Phase 1 extract exists today) and remaining Terraform hardening: NetworkPolicies, PDBs, HPA on a custom RPS metric.
Storefront	A cart and checkout on the shop front. order-service's cart API and crash-safe saga have existed since Phase 3; nothing customer-facing calls them.

KNOWN LIMITATIONS, STATED PLAINLY

- Secrets are plaintext in Git, except the enquiry encryption key (ADR 0021), which is created by hand.
- Single-node cluster: PDBs, anti-affinity, node drains and multi-AZ behaviour cannot be exercised.
- Only one environment is materialised at a time; none has run concurrently with another.
- The AWS layer has never been applied. Module wiring is proven; AWS behaviour is not.
- No sustained-load figures exist, in or out of k3d — only single-threaded smoke-test numbers.
- platform and observability have never run and have no manifests. WSL2 now has ~11 GB (applied by 22 Sep), so observability's ~9.2 GB estimate fits on paper; it is still an estimate.
- The storefront's premium behaviour runs in Chromium only, and has never been seen on a real phone.
- payment-service's acquirer is a stub: no partial captures, chargebacks, 3-D Secure or settlement.
- aquatics-advisor has no HTTP-layer or catalog-client tests, and ships on python:3.11-slim rather than distroless.
- The advisor's predation rule uses adult length, not mouth gape (the catalog does not record it) — stated in rules.yaml itself.